

COMPACT TORI *

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The design of conventional toroidal fusion reactors is constrained by the need to link the plasma torus (and its surrounding blanket) with a set of toroidal-field-generating coils. Compact-torus research is aimed at the objective of creating stable toroidal plasma entities in the absence of an external toroidal field, so that the conventional constraint on reactor design can be removed. Two basic types of grossly stable compact tori have been demonstrated experimentally and theoretically: (1) A low-aspect-ratio toroid of oblate spheroidal outer contour, with finite internal toroidal field (commonly called “spheromak”) can be formed by a “magnetized” coaxial plasma gun, or by a reverse-polooidal-field pinch, or by the transformer action of a “magnetic flux core.” (2) A plasma toroid with a prolate (racetrack-shaped) confining polooidal field and strictly null toroidal field can be formed by dynamic reverse-polooidal-field pinch techniques.

1. Introduction

In fusion reactors of conventional design, the plasma-confinement region tends to be locked into a fixed position by the structure of the external magnetic field coils. In the case of the tokamak, the plasma locations in major radius and axial coordinate are somewhat movable, but the plasma is topologically trapped by the toroidal-field coils. In stellarators and tandem mirror machines, where the external field structure has strong gradients, the plasma location is even more rigidly prescribed.

The concept of a “free” fusion plasma entity, confined by a local magnetic field structure that can move along with the plasma within a uniform external field, has a number of fundamental attractions:

(1) A free plasma entity can be formed in a low-radiation environment and then moved to a reactor section that is particularly well-suited to high neutron fluences and frequent wall-replacement.

(2) A free plasma entity can be brought to ignition temperature in a section specialized to plasma heating (adiabatic compression, injection, rf-heating, etc.) and moved along, after it is ignited.

(3) According to the virial theorem, the equilibrium of a free plasma entity requires the pres-

ence of an externally generated magnetic field component to support the total outward magnetic force; the local magnetic pressure, however, can be maximal at the plasma, diminishing outwards towards the magnet coils. (This is the reverse of the conventional magnetic-confinement situation, where the peak magnetic pressure is typically 4–10 times greater at the external magnet coils than at the plasma.)

(4) While the magnetic field structure of a free plasma entity is expected to decay by resistive diffusion on about the same time scale as the polooidal flux of the tokamak, a reactor based on free plasma entities can achieve quasi-steady-state conditions by keeping a whole sequence of them (an “aging population”) in simultaneous operation. This approach is economically attractive, since the expensive and complex portions of the reactor (i.e., the forming and heating sections) are then exploited in common by a group of reacting plasma entities that spend most of their lifetime generating useful power in a reactor section of relatively simple structure and low cost.

(5) The inherent flexibility of the free-plasma approach can serve to improve reactor performance by allowing for some degree of system redundancy and operator initiative.

The vision of a free-plasma fusion reactor was first entertained in ref. 1. Meanwhile, numerous theoretical and experimental papers have been published in this area: A listing of historical compact-torus work is undertaken in refs. 2 and 3. The

* This work supported by U.S. Department of Energy Contract No. DE-AC02-76CH03073.

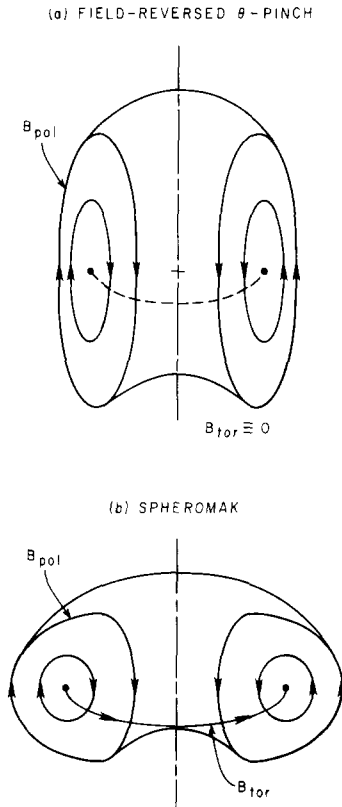


Fig. 1. Two compact tori of principal interest: (a) highly prolate or racetrack-shaped null- B_t case; (b) oblate finite- B_t case (spheromak).

two main lines of approach are illustrated in fig. 1. A “free” plasma, is necessarily a toroid with currents circulating inside it to generate the confining magnetic field. The simplest toroid (fig. 1a) has only a toroidally directed current and a poloidal magnetic field. The average plasma pressure is roughly equal to the magnetic field pressure (“ $\langle \beta \rangle \sim 1$ ”). From the point of view of MHD stability, the second-simplest toroid, which has finite toroidal field (fig. 1b), appears at first sight to be a less daring objective, since beta can be arbitrarily small. This is the case that was considered in ref. 1.

The title of the present paper “Compact Tori” refers to the two basic free-plasma configurations of fig. 1 (not to small-sized tokamaks, etc.). The present paper also limits itself to ordinary plasmas, as distinct from entities with high-energy particle currents. It is one of a roughly contemporaneous set of five (the other four being refs. 4–7) that address this general subject matter. By

arrangement with the other authors, the present paper will seek to emphasize MHD stability, with particular reference to the configuration of fig. 1b, which has come to be called the “spheromak”.

2. The spheromak

A toroid of form (1b) can have null plasma pressure ($\beta \equiv 0$). The special interest of such “force-free” equilibria was first pointed out in the astrophysical context [8]. The stability analysis of such configurations was undertaken in refs. 2 and 9. The stability of a suitably optimized spheromak was found to be of roughly the same quality as that of the reversed-toroidal-field pinch and the tokamak: Gross ideal-MHD stability is insured by a loose-fitting conducting shell. Like the null-beta reversed-toroidal-field pinch, the null-beta spheromak also has the distinction of belonging to the special class of minimum-energy states discussed in refs. 10–12, which can be broadly stable against resistive modes. Gross resistive-MHD stability, however, requires a tighter-fitting shell than ideal-MHD stability. Pressure-driven ideal-MHD modes are stable up to appreciable beta values, but like the pinch, the spheromak is theoretically unstable against resistive pressure-driven MHD modes.

2.1. Stability theory

The ideal MHD theory of the spheromak has been investigated thoroughly by computational means [13,14]. The results are consistent with each other and with the preliminary finding of ref. 2, which addressed a narrower class of configurations. The optimum beta limits reported in refs. 13 and 14 are substantially more favorable than had been foreseen in ref. 2.

The two grossest instability modes of the spheromak correspond to simple tilting and positional shifting ($m = \pm 1$, $n = \pm 1$). The tilting mode is always at least marginally unstable, unless the plasma is oblate and there is at least a loose-fitting conducting shell [2]. In the absence of a conducting shell, the shifting mode is also at least marginally unstable against either horizontal or vertical displacement. Let the poloidal cross section be defined as in fig. 2, with oblateness measured by the parameter ε and with a central “flux hole”

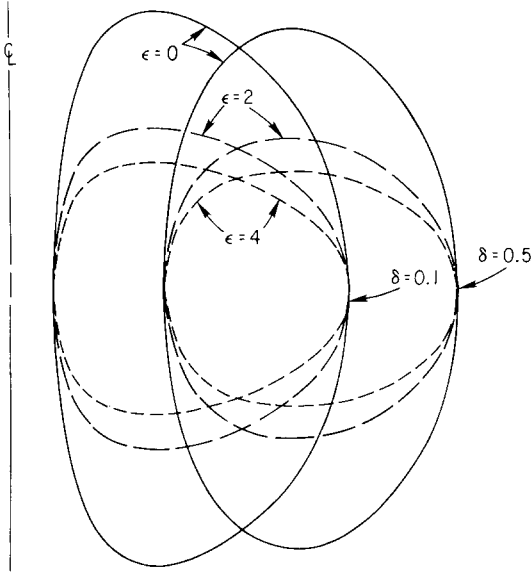


Fig. 2. The external contour of the spheromak is defined by the magnitude of the central flux-hole (δ) and the degree of oblateness (ϵ).

measured by the parameter δ . The plasma is surrounded by the conducting shell defined in fig. 3 [15]. Stability against both tilting and shifting can be obtained if the wall separation is not too great. For an ϵ -value around 2 – not so small as to allow tilting nor so large as to cause horizontal shifting – the critical wall-separation distances ($a_w = b_w$) prescribed by fig. 4 are seen to be quite comfortable.

Satisfactory stability can also be obtained without enclosing the plasma within a complete shell: either an open cylinder ($b_w = \infty$) or a set of

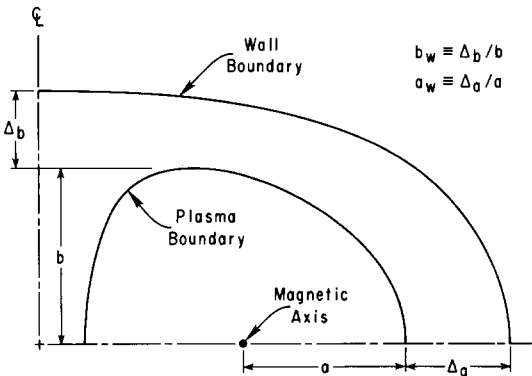


Fig. 3. The separation between the conducting external wall and the plasma boundary is measured by a_w and b_w .

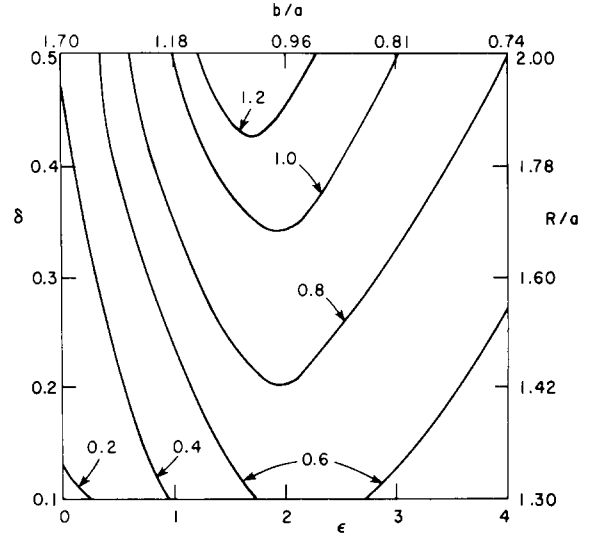
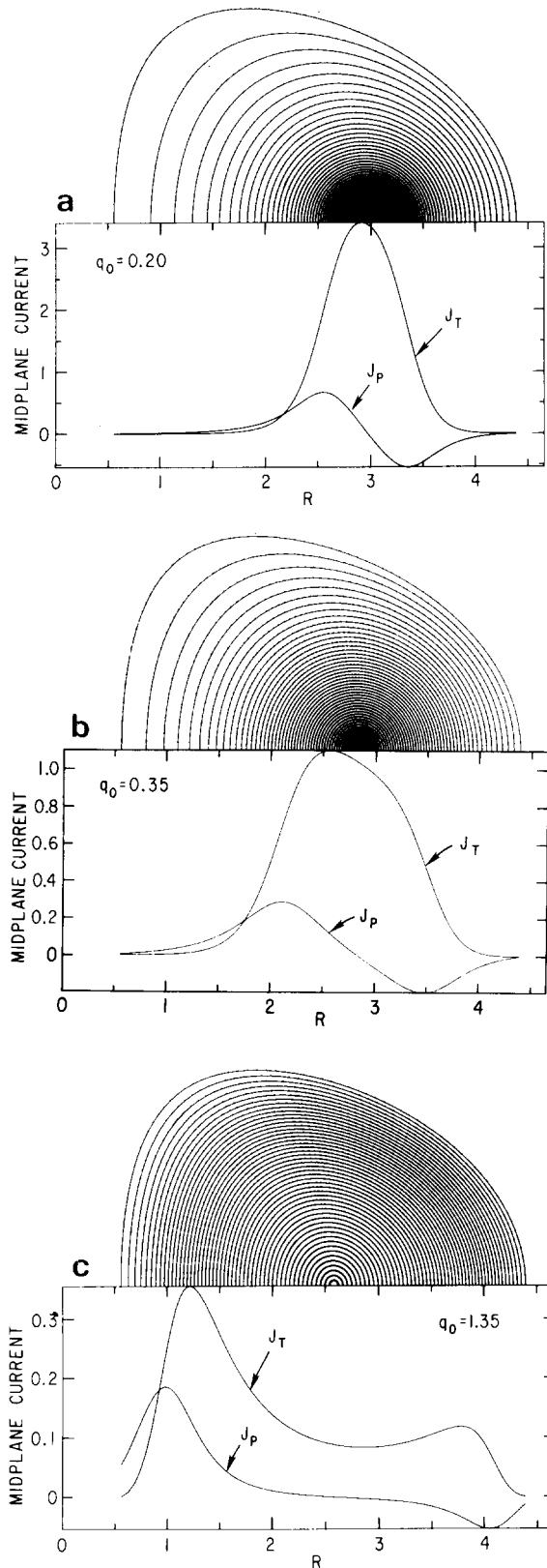


Fig. 4. Stability of the spheromak against tilting and shifting for various values of δ and ϵ . The curves are labeled by $a_w = b_w$; the regions above them are stable.

horizontal plates ($a_w = \infty$) will do. (The latter can be replaced conveniently by pairs of figure-eight loops [16].) If the conductivity of plasma outside the separatrix of the spheromak is taken into account, the effect of line-tying allows the external stabilizing shell to be removed even further from the plasma surface [17].

The stability of the spheromak against internal kink modes and external kink modes with $n > 1$ is quite sensitive to the details of the current distribution. Three characteristic types of spheromak equilibrium are illustrated in fig. 5 for the parameters $\epsilon = 2$, $\delta = 0.1$ (i.e., $R/a = 1.3$), and $\beta = 0$; the central safety factors $q_0 = 0.2$, 0.35 , and 1.35 are used, respectively, with profiles $q(r) = q_0(1 - r^2)$. Case (a) could be characterized as “toroidal-flux-poor”. It has a sufficiently strongly peaked current profile so that the pitch ($qR/a \equiv rB_z/aB_p$) falls well below $2/3$ at $r = 0$. In this range of pitch, there tend to be unstable current-driven internal $m = 1$ kink modes with n -values such that $m/n \geq q_0$; for the specific aspect ratio of case (a), the $n = 3$ mode is predominant. The stability of these internal modes deteriorates if the central shear and/or the oblateness are weak. At the opposite end of the configurational spectrum is the “toroidal-field-rich” configuration of case (c), which has a sufficiently hollow current profile so



that q_0 rises above the inverse of the Kruskal-Shafranov limit $q_0 < 1$. At this stability boundary, the finiteness of the toroidal aspect ratio leads to current-driven internal $m = 1$, $n = 1$ kink modes. For large aspect ratio, where q_0 tends to become small, the dominant stability problem on the high-pitch side is caused by external $m = 1$, $n \geq 2$ pressure-driven modes that are affected adversely by insufficient closeness of the outer shell. Case (b) of fig. 5 lies at the low-toroidal-flux edge of a class of middle-of-the-road spheromaks that can be simultaneously stable against both low-pitch (low- q_0) and high-pitch (high- q_0) kink mode – even if beta is not small – provided that the conducting shell is not too far removed from the plasma surface.

In general, the beta value of the spheromak is limited by the Mercier criterion. (When Mercier is satisfied, the ballooning mode criterion turns out to be satisfied as well.) The limiting beta rises roughly linearly with the pitch $q_0(R/a)$; for moderate aspect ratios, the highest beta is reached at q_0 -values just below unity [13]. For large aspect ratios, beta rises until it is limited by kink modes at an average poloidal beta value over 50%. A spheromak of conventional shape ($\epsilon = 2$, $\delta = 0.1$, $q_0 \sim 0.9$) is found to be stable against all modes up to $\langle \beta_p \rangle = 8\pi \langle p \rangle / \langle B_p^2 \rangle \sim 30\%$, which corresponds to $\langle \beta \rangle = 8\pi \langle p \rangle / B^2 \sim 15\%$ – provided that the wall separations a_w and b_w do not exceed 0.5. Still higher beta values are favored by moderately strong oblateness, and very close-fitting walls.

A peculiarity of the low-pitch internal kink mode in the toroidal-flux-poor spheromak of type (5a) is that the resonant q -value (e.g., $q = 1/3$) need not appear anywhere within the radial profile; hence there is no magnetic island – only a helical perturbation that saturates at small nonlinear amplitude [18]. This phenomenon appears to be closely analogous to the helical equilibrium state of a toroidal-flux-poor reversed-toroidal-field pinch [11]. The high- q_0 kink modes found in the “toroidal-flux-rich” configuration of type (5c), on the other hand, are accompanied by resonant surfaces, and thus may lend themselves to disruption.

Fig. 5 (a). Unstable “toroidal-flux-poor” force-free spheromak model with $\delta = 0.1$, $\epsilon = 2$, $q_0 = 0.20$. (b) Force-free spheromak model with $\delta = 0.1$, $\epsilon = 2$, $q_0 = 0.35$, at the lower edge of the region of stable q_0 period (c) Unstable “toroidal-flux-rich” force-free spheromak model, with $\delta = 0.1$, $\epsilon = 2$, $q_0 = 1.35$.

tive flux-reconnections like those of the tokamak.

The preceding discussion was focussed on ideal-MHD modes. Finite resistivity should not have an important effect on the low- q_0 internal modes, where there is no resonance, but should have an adverse effect on resonant high- q_0 modes, especially surface modes. A preliminary survey of resistive kink instability [2,9] indicates that stabilization may require the external shell to be brought substantially closer to the plasma. In the context of suppressing external modes of this type, low aspect ratio should be valuable – both in enhancing stability and in increasing the ratio of the wall separation to the overall plasma size.

Resistive kink stabilization is particularly difficult for high- n modes; on the other hand, these are the modes most likely to be suppressed by gently graded profiles and by the “gyro-viscosity” of the plasma. Similarly, the short-wave resistive pressure-driven modes are all unstable in principle, and one must look to kinetic effects to limit the damage. Very little seems to be known as yet – either experimentally or theoretically – about the impact of any of these fine-grained resistive instabilities on the actual confinement of high-temperature plasmas.

2.2. Experiments

The creation of a separated toroidal plasma entity requires reconnection of poloidal magnetic flux. In the case of the spheromak, a further requirement is the entrapment of the desired toroidal flux within the separated poloidal-flux region. Thus far, the coaxial-gun-ejection method suggested in ref. 1 (fig. 6) has proved to be most productive in generating experimental results. Gun-produced spheromak plasmas are being studied currently at the Los Alamos National Laboratory (LANL) [4,6,19], and were studied until recently at the Lawrence Livermore National Laboratory (LLNL) [20]. Another convenient spheromak formation method is to use an axial plasma current to generate toroidal flux within the closed poloidal flux region of a reversed-poloidal-field pinch. This approach is being studied currently at the University of Maryland [5,7]. These two familiar spheromak-formation methods benefit from dynamic effects – in the sense that plasma inertia is used to assist in the flux-reconnection process. Recent experiments in both areas

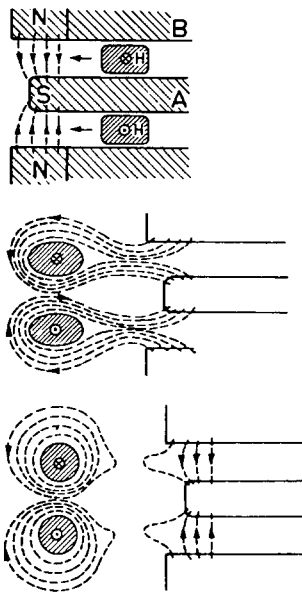


Fig. 6. Coaxial rail-gun method of spheromak formation, first described in ref. 1.

have shown, however, that reconnection can also be achieved quasi-statically when the formation cycle is slowed down. An evolution to quasi-static formation is, of course, desirable from the technological point of view, since the reactor application calls for the storage of very large magnetic energies ($\geq 10^9$ J) in the spheromak confining field [2]: As in the case of other magnetic-fusion approaches, such large energies are delivered most conveniently on a time scale that is suitable for rotating electrical equipment.

The problem of optimal quasi-static spheromak formation is being studied at the Princeton Plasma Physics Laboratory (PPPL) [21]. In order to facilitate the minimization of impurity evolution, an added constraint has been imposed: that the plasma currents should (to the maximum possible extent) remain closed within the spheromak configuration during the formation process, rather than flowing through external electrodes. These considerations have led to the development of a new formation method, using a toroidal “flux core” to induce the desired currents – a method that is technically complex, but appears to provide the desired performance.

From the point of view of topological procedure, the three formation methods currently in use are closely related. The parameters of the initial

experiments have been fairly comparable, with the gun approach yielding the highest toroidal currents (500 kA) and life times (800 μ s), and the flux-core approach permitting the highest electron temperatures (40 eV) to be reached. Some further particulars concerning these experiments are reported in following three sections.

2.2.1. Coaxial-gun injection

As illustrated in fig. 6, a coaxial plasma rail gun is readily adapted to the generation of spheromaks: If an initial poloidal flux in the form of a radial fringe field is placed at the gun muzzle, the gun-accelerated toroidal-flux-bearing plasma ring becomes wrapped in the poloidal flux, which stretches and finally reconnects as the plasma leaves the gun. The experiments at LANL and LLNL use modern high-powered plasma guns with air-core poloidal-field coils, but the basic mechanism is the same: Separated spheromaks of the desired shape are readily produced – even though the gun formation process tends to impart much too high a ratio of toroidal to poloidal flux. Already in the initial experiments [22], there was

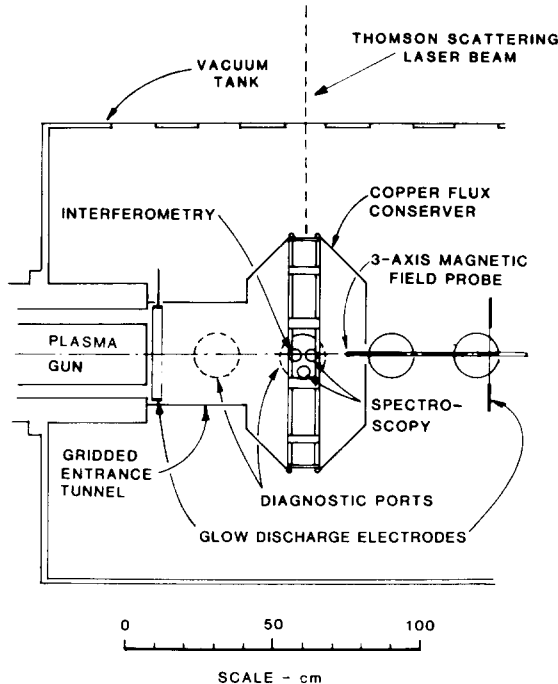


Fig. 7. CTX coaxial-gun facility at LANL injects spheromak plasmas into conducting shells of various shapes.

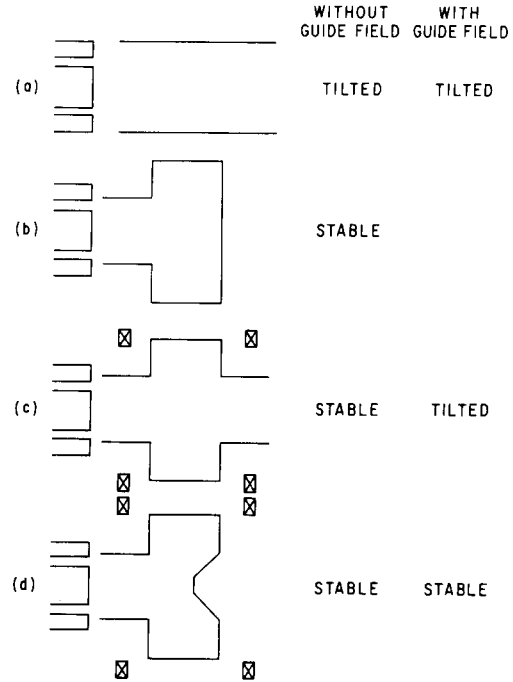


Fig. 8. Spheromak plasmas injected into prolate CTX shells are always tilt-unstable (a). Oblate shells insure stability in the absence of external guide field (b), but there is a threshold for tilting at finite guide field (c). Indentation of the shell (d) enhances tilt stability.

clear evidence that the plasma toroid manages to readjust the proportion of the toroidal and poloidal fluxes by spontaneous internal reconnection processes. This spontaneous self-improvement mechanism of the spheromak is, of course, consistent with the high theoretical expectations that attach themselves to a “lowest energy state” in the sense of refs. 10–12.

The work reported in ref. 22 was based on “fly-by” diagnostics. In the LANL experiments [19] (fig. 7), the spheromak plasma is shot into conducting shells of various shapes (fig. 8) and is thereby brought to a rest, so that more extensive diagnostic studies can be carried out. When an axially elongated shell is used (fig. 8a), so that the spheromak assumes a prolate form, the predicted tilting mode occurs (cf. section 2.1), whether an external guide field is superimposed or not. (When there is no guide field, the poloidal-field pressure of the spheromak is supported directly by means of eddy currents in the conducting shell.) In the null-guide-field case, the spheromak tilts until its former axis of symmetry is transverse to the axis

of the shell (the predicted lowest-energy state) and then decays by resistive diffusion in about $100\ \mu\text{s}$. If there is an appreciable guide field, so that the poloidal flux of the spheromak can reconnect to open field lines as the plasma tilts, the resistive decay is much more rapid. When an oblate shell is used (fig. 8b), the spheromak is stable against tilting and decays slowly by resistive diffusion. If a sufficiently strong guide field (c) is added, however, so that there is an appreciable gap between the last closed poloidal flux surface and the shell, the plasma becomes tilt-unstable again. All these features are in good agreement with the predictions of simple MHD theory (section 2.1). Improved experimental stability against tilting is found to be provided by the indented shell of fig. 8d.

The typical operating mode of the LANL experiment employs two stabilizing half-shells (fig. 7), so that observations can be carried out readily in the mid-plane. (The effectiveness of the stabilizing shells is found to depend on a set of "current bridges" linking them, as shown in the figure.) Some magnetic-field measurements obtained during an early stage of the LANL program are shown in fig. 9. During the first $20\ \mu\text{s}$, the poloidal field of the spheromak is seen to be passing through the entrance region of the shell. After $50\ \mu\text{s}$, the spheromak flux has become completely detached from the flux of the plasma gun. The stably trapped spheromak configuration then decays in another $100\ \mu\text{s}$.

The observed decay rate appears to be consistent with classical expectations for a plasma with (measured) electron temperatures that are typically below $10\ \text{eV}$. The plasma density starts at $2\text{--}4 \times 10^{14}\ \text{cm}^{-3}$ and decays in time to a few times $10^{13}\ \text{cm}^{-3}$. The peak magnetic field strength is of order $0.5\ \text{T}$.

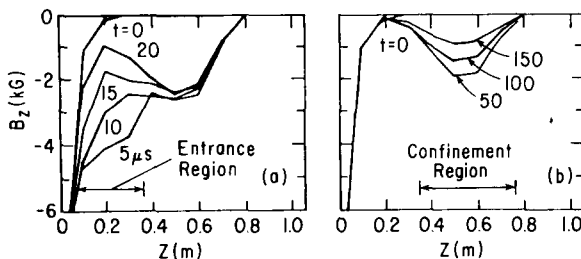


Fig. 9. Magnetic probe measurements carried out in an early phase of the CTX gun-injection program show spheromak trapping and $100\text{--}\mu\text{s}$ lifetime.

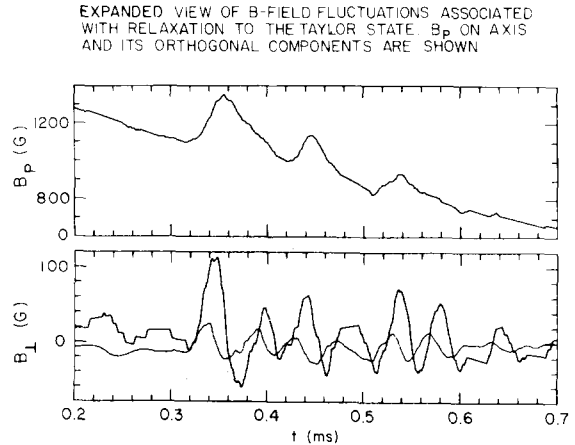


Fig. 10. During the $800\text{--}\mu\text{s}$ decay of the CTX spheromak plasma, distinct increases of poloidal flux take place, which are accompanied by decreases of toroidal flux and nonsymmetric magnetic fields on axis.

In recent LANL experiments [6], much slower decay times have been achieved as a result of intensive impurity-control measures. Fig. 10 shows part of a field-decay process that lasts $800\ \mu\text{s}$. The poloidal-field trace of fig. 10 also exhibits an extremely interesting plasma-physical phenomenon that is common in these experiments – though usually of smaller amplitude: the sudden increase of poloidal flux. Fig. 10 includes two magnetic field traces measured on the axis of symmetry in orthogonal transverse directions. These traces (which should be identically null in a symmetrical configuration) exhibit major asymmetric perturbations coincident with the poloidal flux jumps. When the trapped toroidal flux is monitored by probes inserted into the plasma, negative toroidal flux jumps are seen to accompany positive poloidal ones.

These experimental data provide direct evidence on a controversial topic: the possible nature of the spheromak flux-conversion mechanism. The large-scale conversion of toroidal to poloidal flux that typically takes place during the gun-formation process has been attributed variously to a kink instability at the gun muzzle [22] (fig. 11) and to phenomena connected with plasma rotation [23]. While the first hypothesis might indeed apply during the formation process itself, the data of fig. 10 reveal the existence of a flux-conversion mechanism of a more subtle kind, occurring later in time. Whether this mechanism is of the type proposed in

FLUX ENHANCEMENT BY KINK INSTABILITY

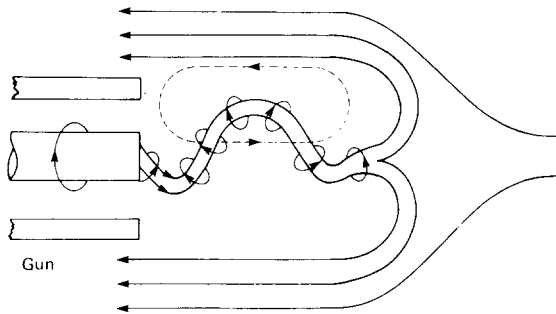


Fig. 11. A possible model for the conversion of toroidal to poloidal flux is kink instability of the plasma column attached to the center conductor of the coaxial gun.

ref. 23 is still debatable. The simplest interpretation is, perhaps, that in a toroidal-flux-rich spheromak configuration like that of fig. 5c, flux-reconnection occurs periodically at the resonant ($q = 1$) surface of the $m = 1, n = 1$ kink mode – in a manner closely analogous to the “sawtooth” activity of the tokamak [24]. In the tokamak, resistive diffusion causes $q(0)$ to fall below unity and the instability then produces a positive Δq , decreasing the poloidal flux. In the spheromak, the same mechanism may well be running in the opposite direction. These recent measurements will clearly be of exceptional interest to devotees of the “lowest energy state” [10–12].

2.2.2. Reversed-poloidal-field pinch with finite B_z

Spheromak formation experiments using theta-pinch techniques have been carried out at the University of Maryland [5,7] in the PS-1 device. As illustrated in fig. 12a, the basic approach is to establish an initial axial field (in the 0.1-T range), then pass an axial current between the tubular electrodes to generate trapped toroidal field, and finally pulse on a reverse axial field (~ 0.2 T, with a rise time of a few microseconds) to create the desired region of closed poloidal flux. Reconnection takes place at each end, and is encouraged by split coils (fig. 12b). The resultant spheromaks have high density (10^{15} cm $^{-3}$) and initially also have high ion temperature (≤ 60 eV) and high beta values [25]. As in the gun experiments, anomalous enhancement of poloidal flux is observed, but individual flux jumps have not been

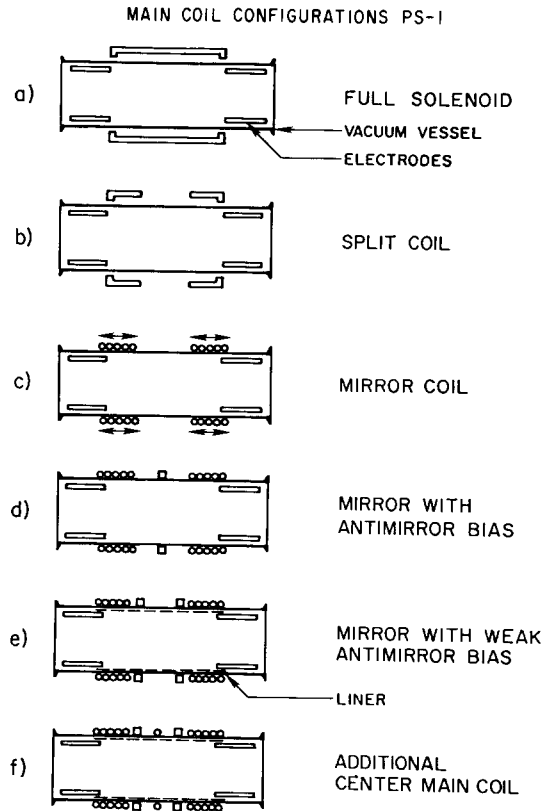


Fig. 12. The reversed-poloidal-field pinch at the University of Maryland is equipped with hollow electrodes for the generation of axial plasma current and toroidal magnetic field. Various coil configurations are used to control stability.

resolved. The plasma lifetimes range up to 30 μ s in tilt-stable cases. The tilting instability is observed in spheromaks of strongly prolate shape, but does not appear to be quite as sensitive to mild prolateness and to the presence of an external axial field as in the case of the LANL gun work. Line-tying by external plasma may be playing a more important stabilizing role [17] in the Maryland experiments.

Fig. 12c indicates a recent modification of PS-1, in the direction of a slower, quasi-static formation process (0.4–0.7 T in 6–16 μ s) using multi-turn mirror coils; the “crowbar” time has been extended to 120 μ s. Plasma densities up to 2×10^{15} cm $^{-3}$ are typically accompanied by electron temperatures below 10 eV. Various complications of the formation topology have been explored, including “doublimaks” and the use of an anti-mirror bias coil (fig. 12d) to control oblateness. Stabil-

ity against the tilting and shifting modes (cf. section 2.1) was improved simultaneously by introducing a thin cylindrical stainless steel sheath as a nearby conducting shell (fig. 12e), and by changing the anti-mirror coils so as to moderate the degree of oblateness (fig. 12e and 12f). Further improvements in stability were achieved by applying the figure-eight-coil scheme of ref. 16. Stable spheromak lifetimes of $40\ \mu\text{s}$ are being observed – apparently limited mainly by radiation cooling and ordinary resistive diffusion.

2.2.3. Flux-core formation

Fig. 13 illustrates a quasi-static spheromak formation process that is being developed at Princeton. In order to avoid the use of electrodes, the toroidal flux is stored initially in a toroidal solenoid (the “flux core”) which also contains a poloidal-field coil. The initial poloidal field (fig. 13a) is weakened on the small-major-radius side by an external vertical field. When the toroidal flux is expelled from the core (by reversing the solenoid current), a tubular discharge is formed around the core (fig. 13b). Under the combined influence of the static vertical field and an appropriately programmed reduction and reversal of the applied poloidal field, the plasma tube then reconnects into a spheromak located on the small-major-radius side of the flux core (fig. 13c). The effectiveness of this process in trapping most of the initial toroidal flux inside the final spheromak configuration has been demonstrated by means of various two-dimensional computer simulations [21,26]. A major new experimental device employing this formation technique, the S-1, is currently under construction (see below). The results of a recent simulation of S-1 performance are illustrated in fig. 14.

The flux-core formation scheme has been tested on several small experimental devices [27]: Proto-S-1A and B are $1/6$ the size of S-1 (flux-core radius $R_c = 15\ \text{cm}$) and have $1/4$ the peak magnetic field strength ($B_t = 0.2\ \text{T}$). Proto-S-1C (fig. 15) is $1/3$ -size ($R_c = 30\ \text{cm}$) and has $2/5$ the peak field strength ($B_t = 0.3\ \text{T}$).

Proto-S-1A and B have confirmed the effectiveness of the basic formation process, producing spheromak plasmas of $10^{15}\ \text{cm}^{-3}$ density, $\sim 15\ \text{eV}$ electron temperature and $20\ \mu\text{s}$ lifetime. The tilting instability was suppressed by applying two figure-eight stabilizing loops [16] (fig. 16) to the two ends

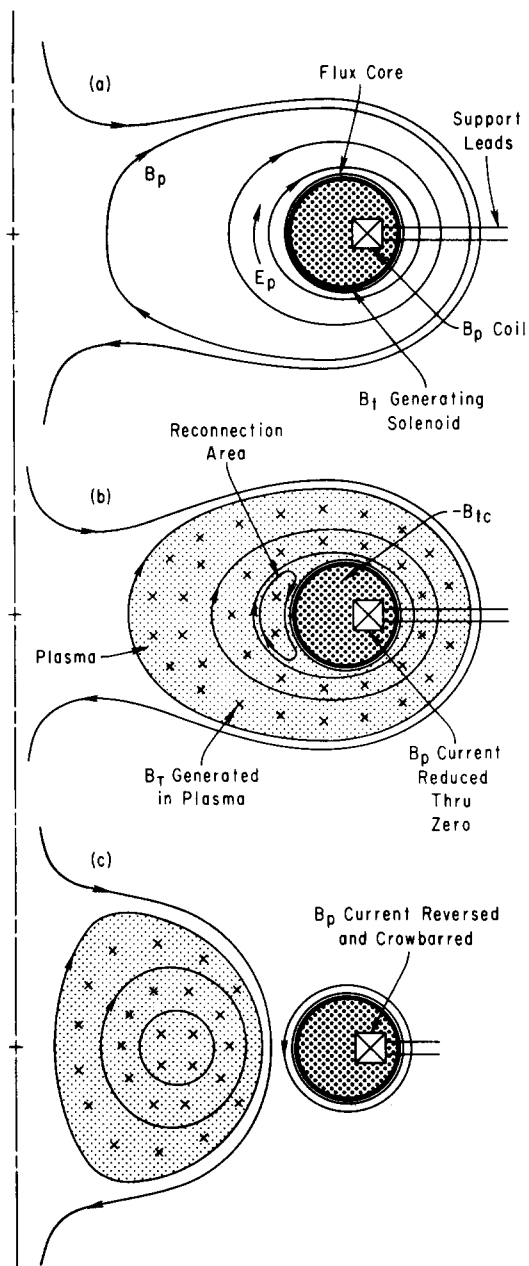


Fig. 13. Quasi-static spheromak formation without use of electrodes is accomplished by means of a toroidal flux core. An initial poloidal field is generated by the B_p coil plus an external vertical field. The plasma is then ionized and a B_t field is induced. Finally, the current in the B_p coil is reversed and crowbarred.

of the spheromak. The tilting mode and its suppression in Proto-S-1B are illustrated in fig. 17, which shows computer-reconstructed constant- B_t contours based on multiple magnetic-probe measurements inside the plasma.

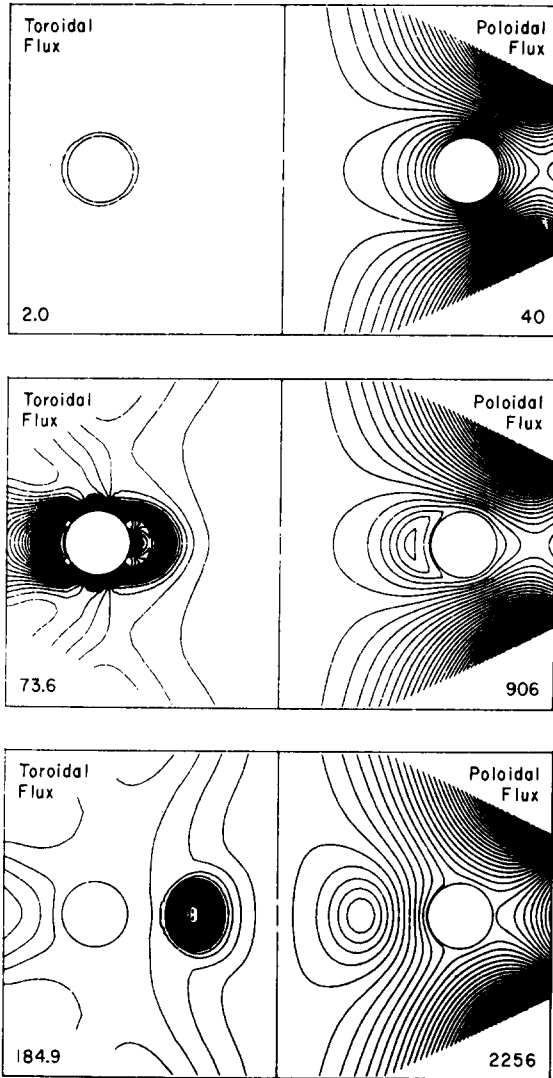


Fig. 14. Computer simulation of the spheromak formation process of fig. 13, for the parameters of the S-1.

Spheromaks with considerably better parameters are being produced in Proto-S-1C. For typical densities of $1\text{--}5 \times 10^{14} \text{ cm}^{-3}$, the electron temperatures (measured by Thomson scattering) range up above 40 eV at the low-density end (fig. 18). Since the Proto-S-1C spheromak is held in an external vertical field, normally well removed from wall-stabilization, tilting and shifting instabilities tend to terminate the plasma lifetime unless special counter-measures are taken. Fig. 19 illustrates a typical case, where the spheromak lives about 70 μs as a separated entity and then terminates abruptly. A particularly effective stabilizing tech-

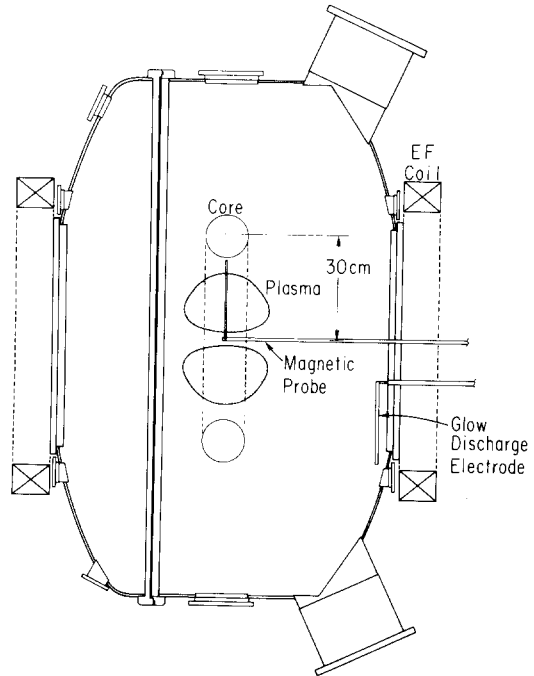
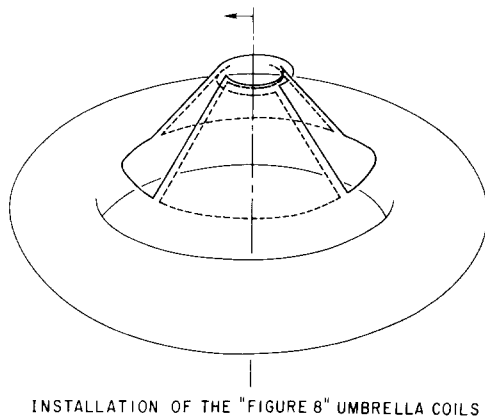


Fig. 15. The proto-S-1C device, a small-scale, low-field model of the S-1.

nique is to insert a copper rod into the central hole of the toroid (fig. 20). In that case, the plasma regime of fig. 19 lasts about twice as long and terminates gently, by resistive diffusion (fig. 21).

While the insertion of the center rod may appear to compromise the principle of the "free plasma entity" to some extent, the rod is not so



INSTALLATION OF THE "FIGURE 8" UMBRELLA COILS

Fig. 16. The stabilization of tilting and shifting modes by means of figure-eight coils [16] was demonstrated on proto-S-1B. The plasma is placed inside the flux core, between two "umbrella coils."

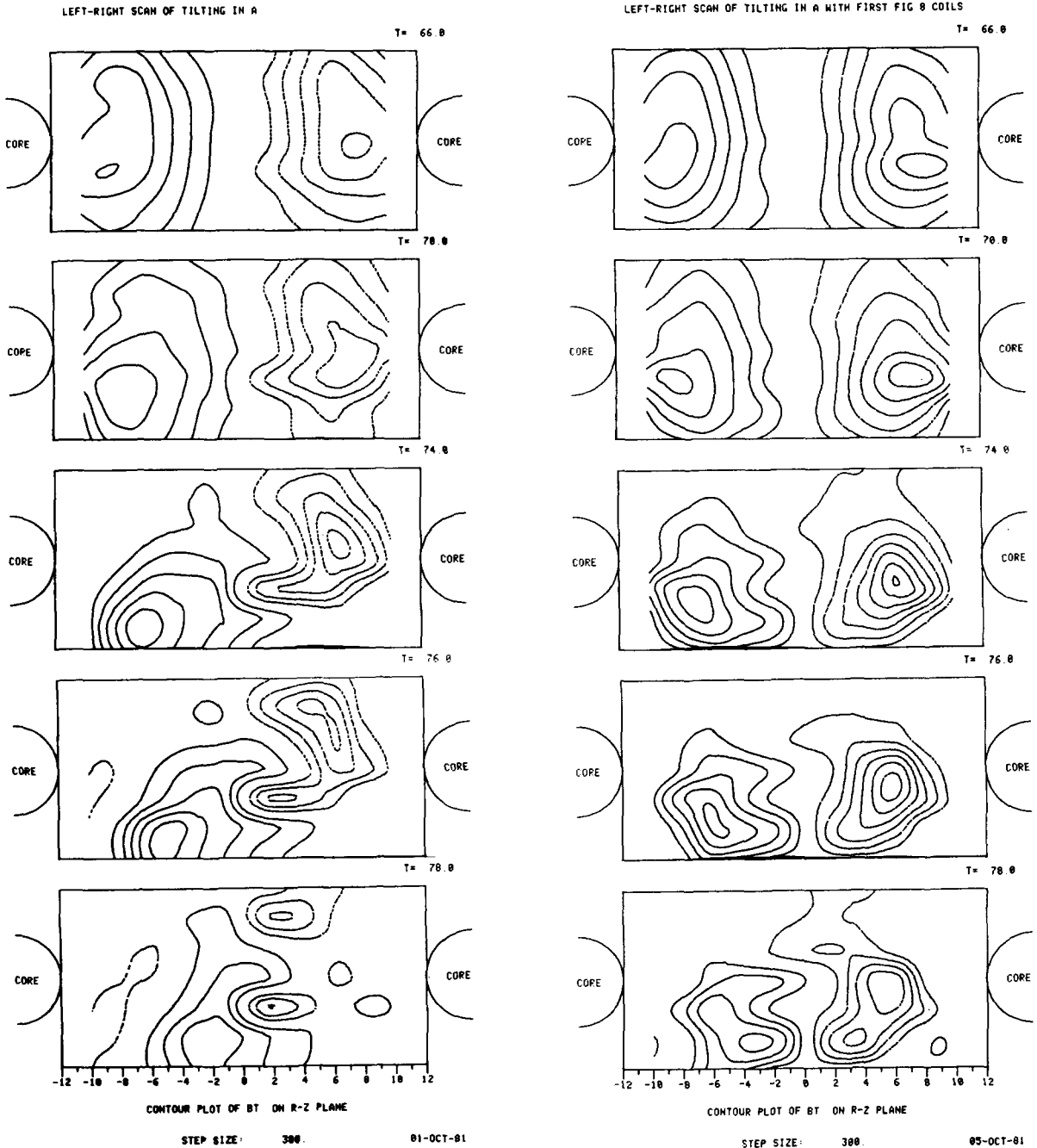


Fig. 17. In the proto-S-1B spheromak experiments, the plasma became tilt-unstable after some $10 \mu\text{s}$. Stabilization by umbrella coils (fig. 16) proved to be effective.

restrictive as a surrounding shell – and, of course, is very much less restrictive than a surrounding toroidal-field-coil system. As illustrated in fig. 22, if the center rod is supported mechanically at one

end, the spheromak can still be moved freely when desired. The concept presented schematically in fig. 22 is that a spheromak at rest in an external guide field needs a kind of “socket” to keep it

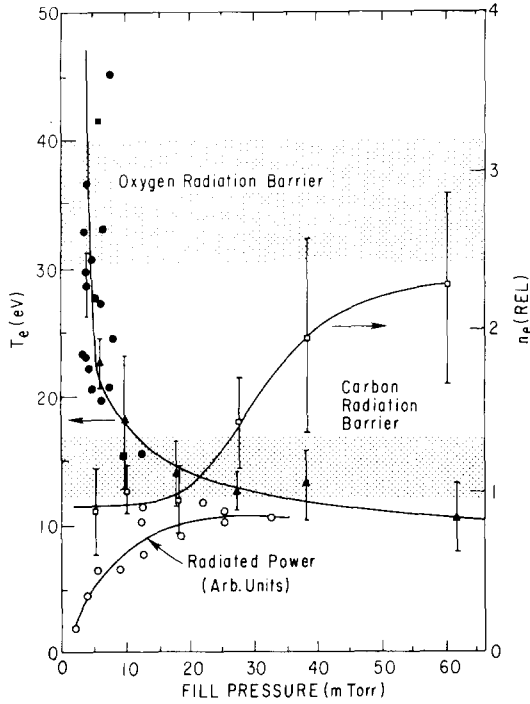


Fig. 18. Thomson-scattering measurements of central T_e in proto-S-1C indicate that the light impurity radiation barrier is beginning to be surmounted at low densities.

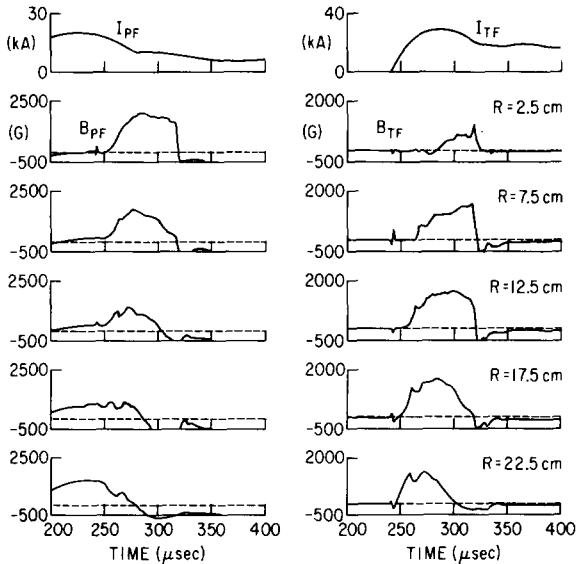


Fig. 19. In the Proto-S-1C experiments, the plasma becomes abruptly unstable against tilting and/or shifting modes, after about 70 μ s.

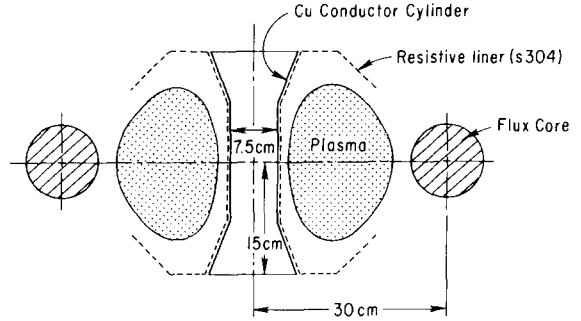


Fig. 20. The proto-S-1C plasma can be stabilized by a central copper conductor. (More complete enclosure by the resistive liner turns out not to be necessary.)

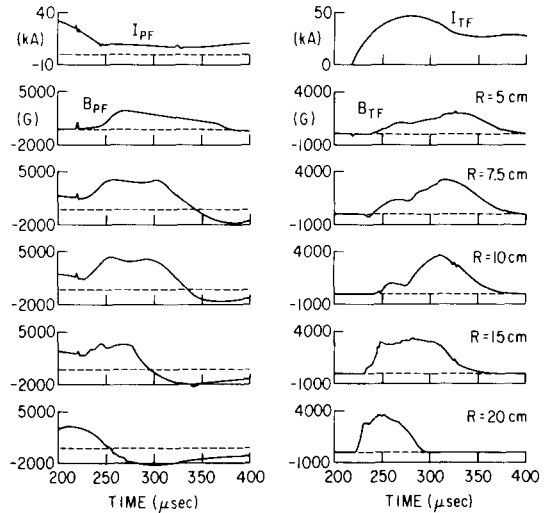


Fig. 21. Stabilization by the central conductor of fig. 20 proved to be effective in suppressing the gross instability shown in fig. 19.

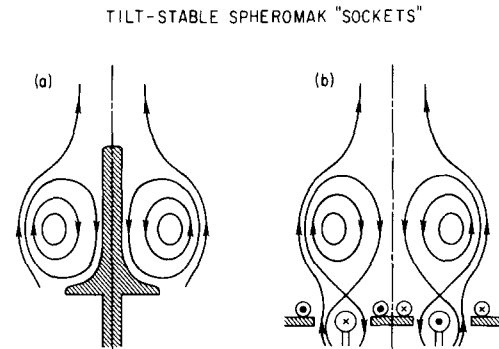


Fig. 22. Since spheromak plasmas tend to be unstable against tilting and shifting when isolated in an external vertical field, the use of "sockets" is appropriate to hold them in place. The central rod (a) appears to be effective; the poloidal-divertor approach (b) is being studied.

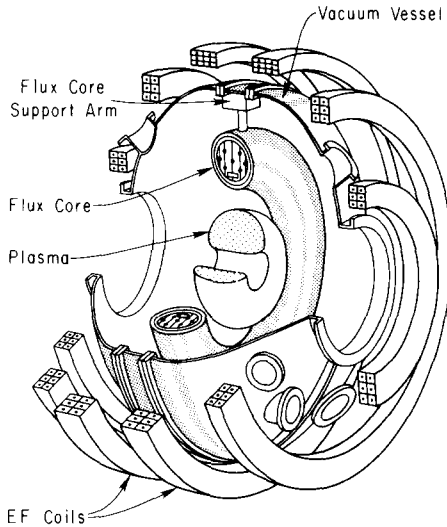


Fig. 23. The S-1 spheromak at PPPL.

from tilting or shifting. There is reason to expect that a poloidal divertor structure (fig. 22b) may constitute an even more satisfactory “socket” than a centering rod. Computer studies of the stability of this configuration are being undertaken, and experimental tests will be carried out on Proto-S-1C.

The S-1 device (figs. 23 and 24) is designed to produce a sufficiently large ($R_c = 1$ m) and hot ($T_e \geq 100$ eV) plasma, so that microscopic confinement in the spheromak configuration can begin to be assessed. During the past four years, consistent experimental and theoretical progress has been made in discovering the formation processes and MHD stability properties of the spheromak. The results to date are already of substantial plasma-physical interest – but the relevance of the spheromak concept to the production of practical fusion power will depend critically on the nature of energy-transport phenomena that have been masked, thus far, by impurity radiation and low-temperature resistive diffusion. If S-1 performance can be inferred from performance in the Proto-S-1 experiments, the plasma should be able to burn through the carbon and oxygen radiation barriers and begin to explore transport-limited high-temperature spheromak confinement regimes.

Construction of the S-1 was undertaken in 1980; completion is expected in the spring of 1983. The initial S-1 flux core will be able to produce oblate spheromak plasmas of 0.4-m major radius and

0.25-m minor radius, with toroidal plasma currents of ~ 0.5 MA. An optimized version of the tilt and shift-stabilizing techniques currently under development in Proto-S-1C will be selected for S-1. The plasma lifetime is expected to be in excess of 1 ms. If the initial confinement results are favorable, hotter, higher-current spheromaks could be produced in S-1 by more powerful flux cores. Alternatively, the S-1 spheromak plasma could be removed along the axial guide field to an adjacent compression-heating or beam-injection section.

3. Reversed-poloidal-field pinch with null B_t

Compact tori of the type illustrated in fig. 1a are reviewed in some detail in refs. 4 and 6. A brief discussion is given here, with emphasis on points of comparison with compact tori of type 1b.

3.1. Stability theory

At first sight, the absence of toroidal field seems likely to have a uniformly deleterious effect on gross stability, but this does not turn out to be the case. Oblate toroids of the $B_t \equiv 0$ type are essentially identical with the simply “dynamic pinch”, which has long been theoretically expected and experimentally shown to have explosively unstable short-wave kink ($m = 1$) and sausage ($m = 0$) modes [28]. Highly prolate null- B_t toroids, however, were discovered experimentally [29] to have impressive stability properties that are slowly gaining theoretical appreciation.

The same MHD calculation that predicts the tilting instability of the spheromak also indicates tilting for the null- B_t toroid; however, experimentally the tilting mode is rarely seen in the latter case. Part of the theoretical explanation is that the poloidal flux surfaces of highly prolate experimental plasmas are not elliptical (fig. 25a) but race-track-shaped (fig. 25b), in which case the tilting mode becomes very weak [30]. Some sort of non-MHD effect appears to be needed in addition, to explain complete stability against tilting – but in view of the presence of field nulls on the separatrix of the null- B_t toroid, the existence of such an effect would be plausible.

The presence of a highly elongated reversed-field layer suggests the possibility of resistive tearing into short-wave rings ($n = 0$); this mode, however,

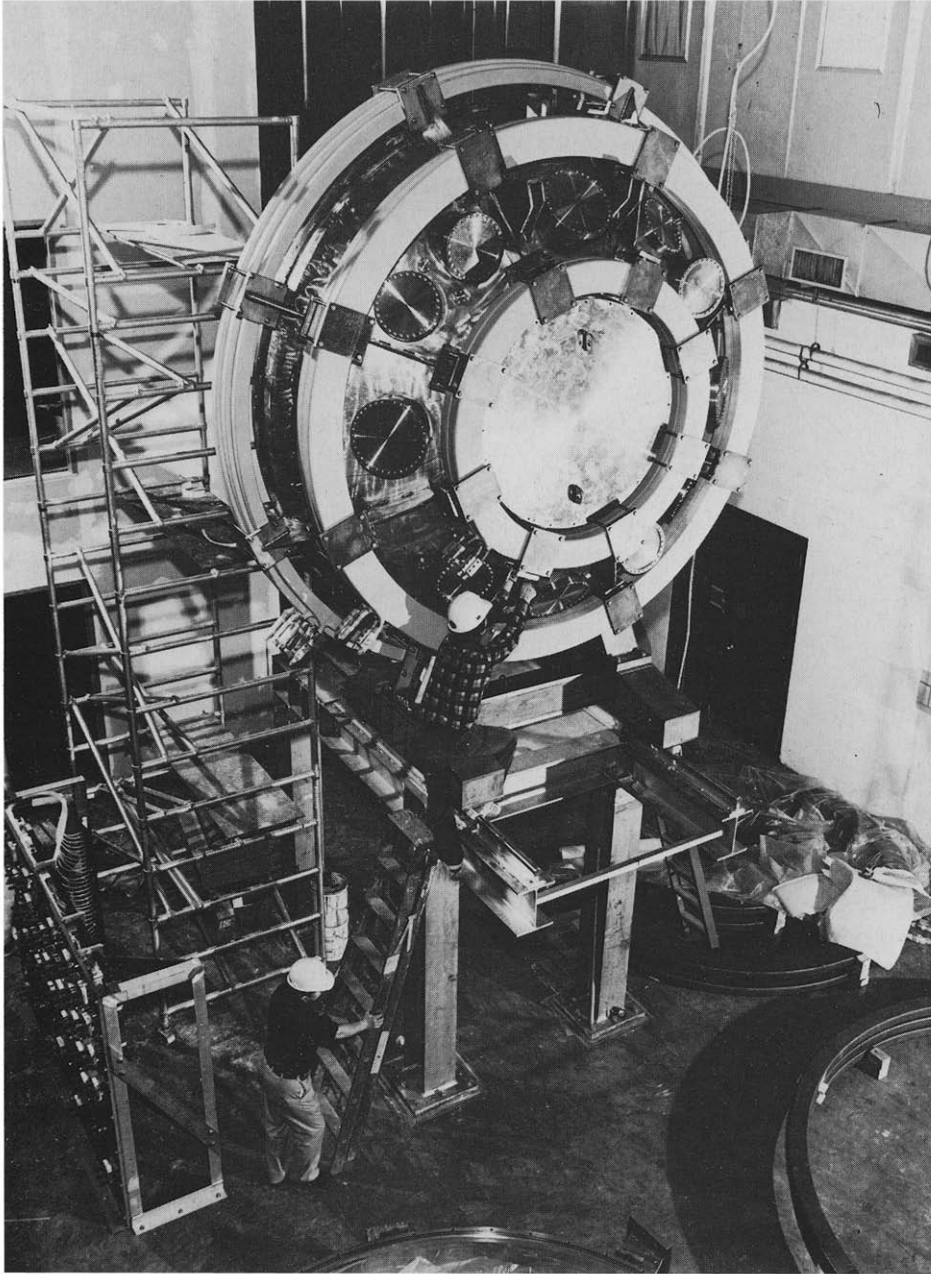
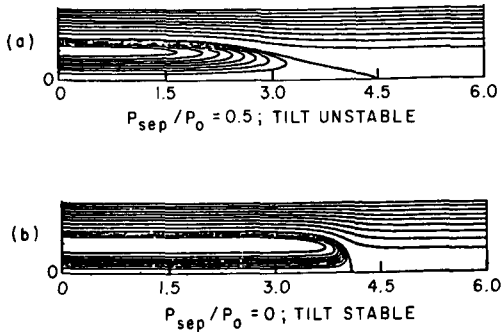


Fig. 24. S-1 construction will be completed in early 1983.

is readily found to be stabilized by the same conducting shell that is needed to establish a two-dimensional equilibrium. In this respect, the null- B_z configuration actually turns out to be much more easily stabilizable than the finite- B_z case, where resistive tearing can occur with its resonant layer $\mathbf{k} \cdot \mathbf{B} = 0$ located on *any* flux surface, rather than

constrained to the $B_p = 0$ surface. (Shifting modes are trivially stabilized by a prolate conducting shell that is sufficiently close to provide equilibrium.)

The most dangerous looking theoretical instability mode of the null- B_z toroid appears to be a residual form of the interchange mode that results



from the adverse poloidal-field curvature at the two ends of the toroid. While the calculated growth rate is weak in highly prolate cases, the nonappearance of this type of instability in the experiments must be ascribed to non-MHD effects. This

Fig. 25. Recent MHD computer simulations [30] of highly elongated null- B_z tori have succeeded in approximating the experimentally observed "racetrack" shape (b). The latter case is considerably more stable than the ordinary prolate case (a) against tilting and flute modes.

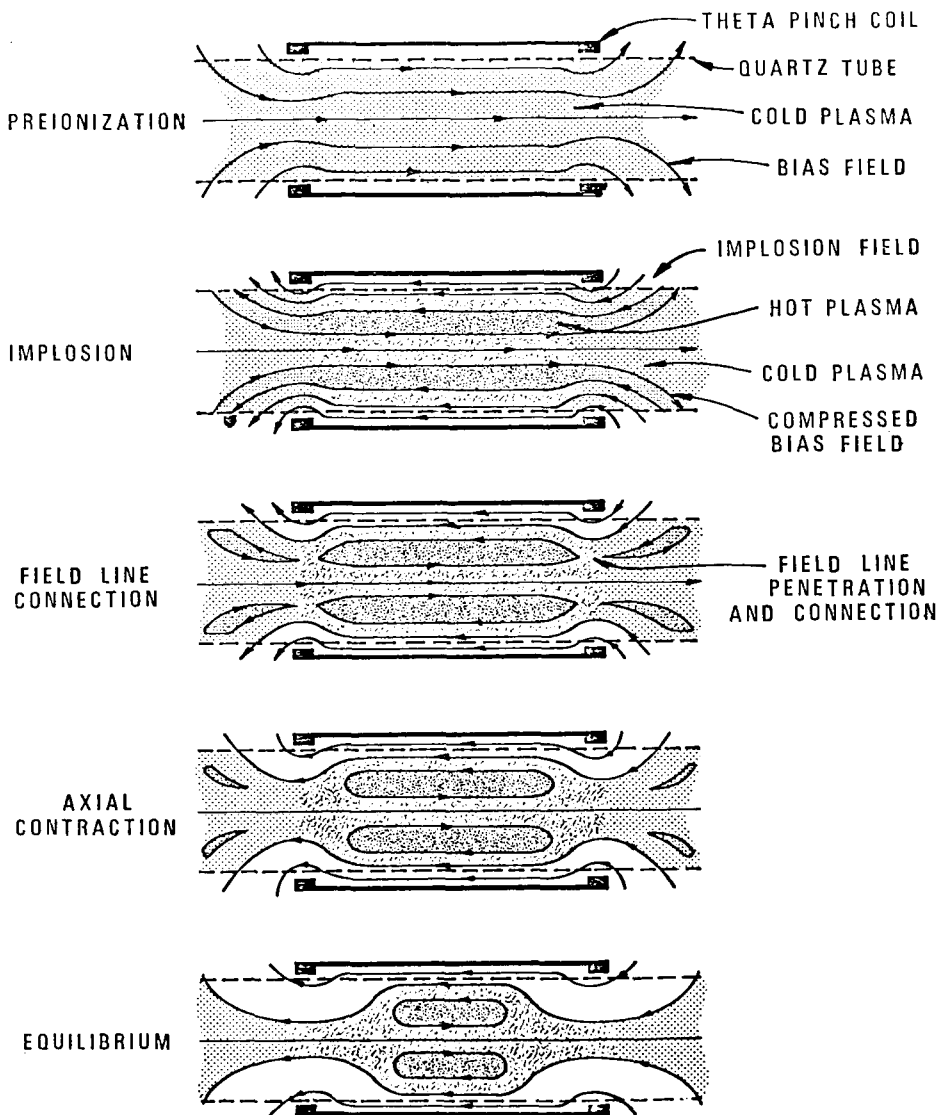


Fig. 26. Formation process of the reversed-poloidal-field pinch.

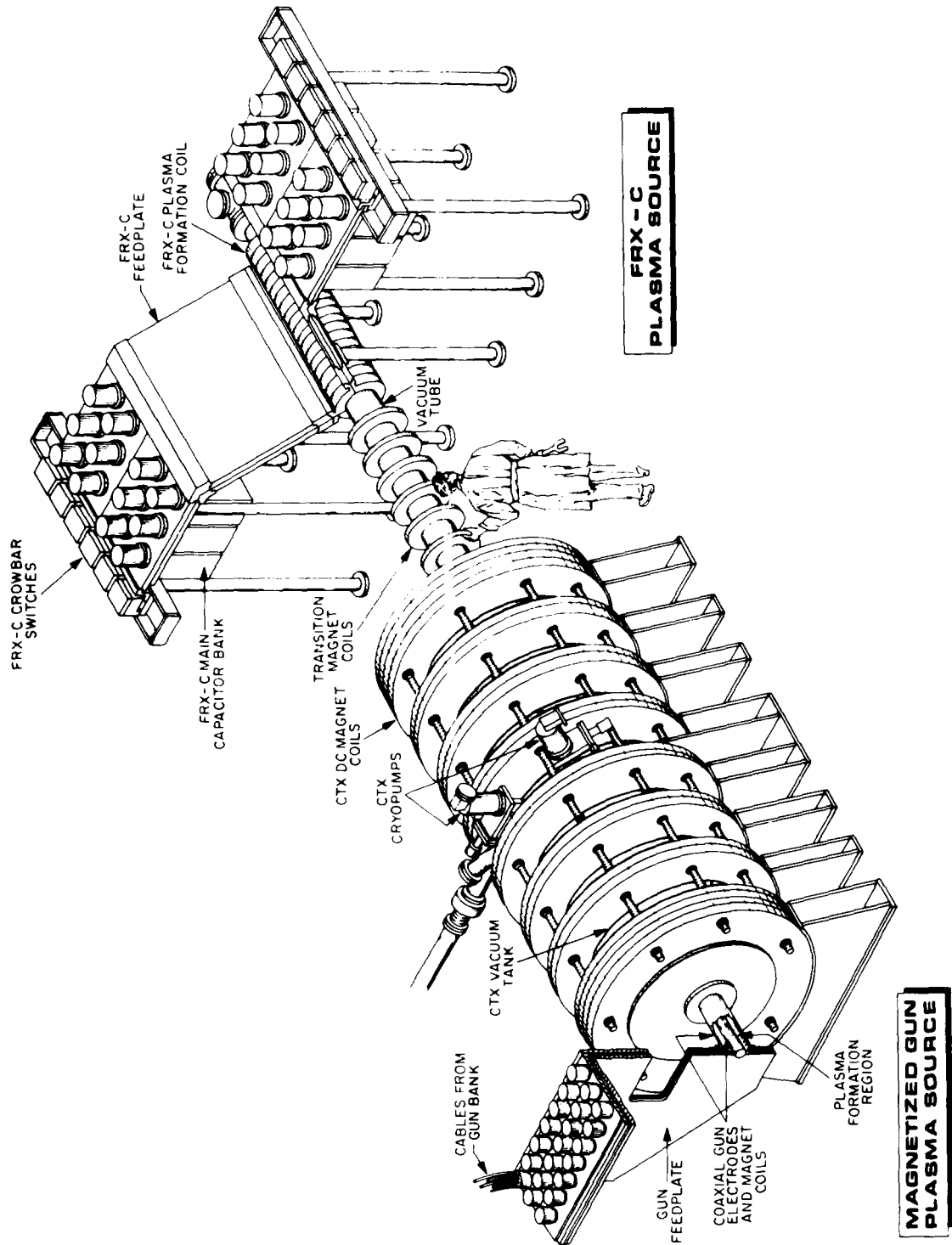


Fig. 27. The CTX tank at LANL, with the spheromak gun-injection experiment at one end and the FRX-C reversed-poloidal-field pinch at the other end.

interpretation places at least a minor cloud on the horizon: In the reactor application, the poloidal flux of a compact torus should encompass numerous ion gyro-radii in order to guard against non-MHD instabilities driven by diamagnetic currents, and to provide a margin of safety for good microscopic particle confinement. If the opposite requirement (i.e., finiteness of the ion gyro-radius) proves to be essential to protect against MHD interchange modes, the "parameter window" for practical reactor operation may turn out to be small or even nonexistent.

3.2. Experiments

The dynamic reversed poloidal-field method [29] is a natural way to produce highly prolate toroids of the type of fig. 1a. Some details of the formation process are illustrated in fig. 26. The basic method is the same as that described in section 2.2.2, but since there is no toroidal field to prevent radial expansion of the plasma at the time of poloidal field reversal, the process is either carried out on a highly dynamic time scale, or else transient external multipole fields must be applied to the plasma surface [29].

The study of null- B_t compact toroids is being pursued at LANL in the FRX-C device, which is located at the opposite end of the previously described spheromak-injection facility (fig. 27). The plasma parameters in the null- B_t case are much more impressive than in the spheromak case: The 1-T axial poloidal field confines plasmas of density $\sim 4 \times 10^{15} \text{ cm}^{-3}$, with $T_e + T_i \sim 400 \text{ eV}$. The toroidal plasma current is 1.5 MA. Since beta is

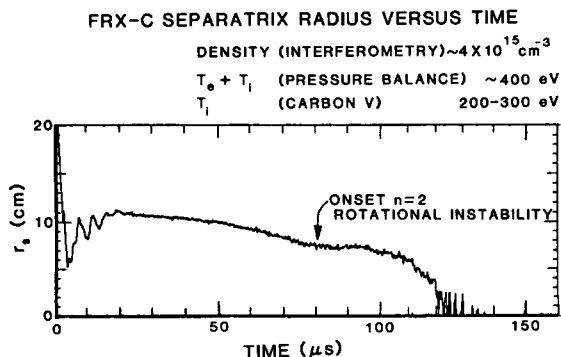


Fig. 28. Reversed-poloidal-field pinches with beta values of order unity have lifetimes in the 100- μs range in the FRX-C experiment.

unity, the energy confinement time and the configurational lifetime are about the same: As illustrated in fig. 27, the toroid decays slowly during a period greater than 100 μs , and is then terminated by a strong instability – a rotational mode that is commonly found in theta pinches. This is a mode that may indeed belong to the class of unstable phenomena thriving on finite-gyro-radius plasma conditions. Within the present range of FRX experiments, however, the measured $n\tau$ -value is still found to increase with increasing number of gyro-radii, and already ranks favorably in the hierarchy of magnetic fusion concepts (just behind tokamaks and stellarators). Many possibilities for further stabilization remain to be explored.

I am indebted to Dr. Stephen Jardin for helpful discussions concerning spheromak formation and stability, and to Dr. Francis Troyon for pointing out the potential advantage of low-aspect-ratio spheromaks in respect to external kink modes. I should also like to thank Drs. Robert Ellis, George Goldenbaum, Rulon Linford, Arthur Sherwood, Richard Siemon and Masaaki Yamada for making available unpublished material for use in this article.

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